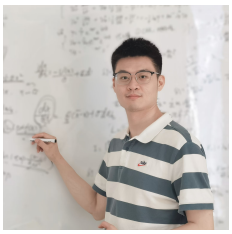

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Current Affiliation

School of Aeronautics, Universidad Politécnica de Madrid. 2026.01 – Present
Postdoc, supervised by Prof. Javier Jiménez.

Education

College of Engineering, Peking University. 2020.09 – 2025.07
Ph.D. in *Fluid Mechanics*, supervised by Prof. Jianjun Tao.

School of Mechanical Engineering, Tianjin University. 2016.09 – 2020.07
B.Eng. in *Ocean Engineering*.

Dept. of Mechanical Engineering, Massachusetts Institute of Technology. 2020.01 – 2020.07
Visiting Student in *Ocean Engineering*, supervised by Dr. Yuming Liu.

Research Interest

My primary research is dedicated to **Turbulence & Transition** ($\mathbf{T^2}$), exploring the underlying physical mechanism with dynamical system approaches, brute-force DNS, experimental data, and machine learning tools.

Publications

* denotes the corresponding author.

Journal Articles

[3] **J. Guan**, H. Hu, T. Hao, H. Wang, Y. Ren*, and D. Fan, "Vortex-induced drag forecast for cylinder in non-uniform inflow," *Physics of Fluids (Letter)*, vol. 37, no. 7, p. 071 708, 2025. Link.

[2] **J. Guan** and J. Tao*, "Distinct lifetime scaling laws of turbulent puff in duct flow," *Physics of Fluids (Letter)*, vol. 37, no. 8, p. 081 703, 2025, [Editor's Pick]. Link.

[1] **J. Guan** and J. Tao*, "The Onset of Metastable Turbulence in Pipe Flow," *arXiv*, 2025. Link.

Conference Proceedings

[3] **J. Guan** and J. Tao*, "Optimal DoS Attacks on Remote State Estimation with Continuous Action Spaces," in *International Congress of Theoretical and Applied Mechanics (ICTAM)*, Daegu, Korea, 2024.

[2] Y. Ren, Z. Duan, **J. Guan**, W. Liu, and Y. Lin, "Optimal DoS Attacks on Remote State Estimation with Continuous Action Spaces," in *2024 IEEE International Conference on Unmanned Systems (ICUS)*, [Best Paper Award], Nanjing, China: IEEE, 2024, pp. 1153–1158.

[1] **J. Guan** et al., "A Study of Fish Undulatory Swimming Based on Merged CFD and Experimental Video of the Swimming Movement of Mosquito Fish," in *OCEANS 2021: San Diego –Porto*, San Diego, CA, USA: IEEE, 2021, pp. 1–6.

Academic Service

- Co-founder of Intelligent and Bio-inspired Mechanics Seminar (IBiM). 2020 – Present
- Reviewer for *Physics of Fluids*.

Honors & Awards

- The Third-prize Scholarship, Peking University. 2022,2021
- Excellent Scientific Research Prize, Peking University 2021
- **Top Ten Outstanding Youth of School of Civil Engineering (0.28%), Tianjin Univeristy. 2019**
- **National Endeavor Scholarship, Ministry of Education of China 2018**